

TalkWeaver: Evidence-Grounded Conversation Maps for Chaotic Multi-Speaker Meetings

TalkWeaver Project

Abstract

Multi-speaker meeting transcription is fragile because overlapping speech, interruptions, ambiguous speaker boundaries, low-energy speech, and domain-specific terms jointly degrade automatic speech recognition (ASR) and downstream correction. The bottleneck is not only recognizing words, but knowing which parts of a fluent transcript are supported by audio, speaker-time evidence, retrieval evidence, or correction audits. We propose TalkWeaver, an evidence-grounded conversation-map framework that treats chaotic meeting transcription as an auditing problem rather than merely an ASR or summarization problem. TalkWeaver combines ASR hypotheses, diarization turns, temporal anchors, overlap indicators, retrieved domain terms, constrained correction proposals, and audit flags into a traceable representation for review. Across existing public-data subsets and controlled safety studies in this repository, meeting speech is substantially harder than read speech, the automatic map runs on English and Mandarin meeting clips, conservative RAG improves base-model Earnings-22 term recall from 0.833 to 1.000 without changing WER on the frozen v3 subset, and leakage audits show that learned correction gates remain unsafe without stronger held-out evidence.

1 Introduction

Modern speech systems can produce fluent transcripts from a single speaker, and large language models (LLMs) can make those transcripts look cleaner. Meetings are a different setting. Speakers interrupt each other, speak over each other, leave partial turns unfinished, use low-energy backchannels, and mention technical or financial terms that are rare in ASR training data. A clean paragraph can therefore hide exactly the information a meeting record should preserve: who spoke, when it happened, whether speech overlapped, which term was retrieved, and whether a correction was supported.

This paper uses the following central framing: chaotic meeting transcription should be treated as an evidence-grounded auditing problem, not merely an ASR or summarization problem. The objective is not to claim state-of-the-art transcription. The objective is to make uncertainty inspectable before a user trusts a corrected meeting record.

Two challenges motivate the design. First, read-speech ASR performance can overstate robustness on meetings. In the repository’s fixed public-data subsets, a base model obtains Mandarin FLEURS CER 0.113, while AISHELL-4 meeting

subsets remain much harder: CER 0.610 on the 12-clip formal subset and CER 0.537 on the 60-clip subset [3, 1]. English AMI meetings also remain difficult, with base WER 0.398 in the formal subset and 0.349 in a 24-clip held-out subset [2]. Second, post-ASR correction can introduce unsupported edits. Retrieval can rescue rare terms, but irrelevant candidates can bias a correction module; learned gating can also look strong when it sees leakage-prone audit fields.

TalkWeaver addresses these challenges by converting speech-system artifacts into a temporal-anchor conversation map. Each map item preserves raw ASR text, timestamps, speaker labels when available, overlap flags, retrieved terms, proposed corrections, and audit outcomes. The Streamlit product and detective-style UI are therefore research artifacts: they expose evidence and uncertainty for inspection rather than serving as the main algorithmic claim.

The contributions are:

- We formulate chaotic meeting transcription as an evidence-auditing task over ASR, diarization, overlap, retrieval, and correction provenance.
- We implement a temporal-anchor conversation map that preserves raw text and traceable evidence for each proposed modification.
- We evaluate the pipeline on existing public subsets of FLEURS, AMI, AISHELL-4, and Earnings-22, while explicitly separating real-public, controlled, proxy, and mock/demo evidence.
- We report both positive and negative results: conservative RAG can improve domain-term recall without transcript drift on a small blind subset, while EvidenceGate leakage and independent held-out failures show the limits of automated correction decisions.

2 Related Work

Multi-speaker ASR and diarization. Multi-speaker meeting processing must connect lexical content to speaker identity and time. DiarizationLM shows that a compact textual representation of ASR and diarization output can be post-processed with LLMs to improve speaker-attributed transcripts [7]. DM-ASR uses diarization as an explicit structural prior for speaker- and time-conditioned ASR queries [5]. TagSpeech proposes end-to-end multi-speaker ASR and diarization with fine-grained temporal grounding [4]. TalkWeaver differs in

scope: it does not train a new speech model or optimize an end-to-end ASR objective. It adapts the speaker-time representation idea into an auditable post-processing layer over existing ASR and diarization outputs.

Overlap-aware meeting processing. Overlap and interruptions are central failure modes in real meetings. A diarization system can give speaker turns and overlap estimates, but overlap F1 can remain low even when DER is usable. In the repository’s AMI held-out diarization result, pyannote obtains DER 0.106 and overlap F1 0.490; on the AISHELL-4 60x20 subset, DER is 0.327 and overlap F1 is 0.262. TalkWeaver uses overlap as a risk signal for correction and review, not as a claim that overlap is solved.

Retrieval-augmented ASR correction. Retrieval-augmented correction has been proposed for rare named-entity ASR errors by retrieving candidate entities and passing them to a correction model [6]. TalkWeaver uses a smaller and more conservative variant: local glossary retrieval supplies candidate domain terms, but a correction can be rejected or routed to review if evidence is weak. This design is motivated by the repository’s RAG v2 negative result, where RAG helped the weaker tiny ASR model but reduced term F1 for the stronger base model.

LLM post-processing and correction safety. LLMs can make ASR text fluent, but fluency is not evidence. TalkWeaver therefore keeps raw transcripts, candidate terms, correction proposals, and gate decisions side by side. The controlled safety experiments and EvidenceGate audit are treated as safety analyses rather than as real-world correction performance.

3 Method

3.1 Problem Formulation

Given an audio recording x , an ASR system emits a sequence of transcript segments $A = \{a_i\}$, where each segment includes text and optional word timestamps. A diarization system emits speaker-time intervals $D = \{d_j\}$. Retrieval over a domain glossary returns candidate terms R_i for segment a_i . A correction module may propose corrected text c_i . TalkWeaver constructs a conversation map $M = \{m_i\}$ in which each item records:

$$m_i = (t_i^{start}, t_i^{end}, s_i, y_i, o_i, R_i, c_i, q_i),$$

where s_i is the attributed speaker or unknown speaker, y_i is raw ASR text, o_i is an overlap indicator, and q_i is an audit state such as accepted, rejected, or needs review. TalkWeaver does not train a speech model and does not introduce a task loss; it organizes model artifacts and correction decisions into a reviewable representation.

3.2 Overall Architecture

Figure 1 summarizes the pipeline. Audio is normalized and passed to ASR. Diarization produces speaker-time evidence. Alignment connects ASR segments or words to speaker turns. Overlap and interruption candidates are derived from temporal speaker evidence. A RAG module retrieves small candidate term lists from a local knowledge base. Correction modules operate under constraints: timestamps and speaker labels are preserved, unsupported changes are audited, and uncertain overlap regions can be routed to review.

3.3 Temporal-Anchor Conversation Map

The conversation map is the central data object. Each item stores start and end time, speaker label, raw text, corrected text if any, overlap status, confidence or review status, retrieved terms, and correction provenance. This follows the temporal-grounding motivation of recent multi-speaker ASR work, but it remains a post-processing format rather than an end-to-end model. In the repository, this behavior is implemented around the alignment, overlap, reference evidence, workflow, and export modules, and it is summarized by the automatic workflow CSV files listed in the claim audit.

3.4 Speaker-Time and Overlap Evidence Extraction

Speaker evidence comes from pyannote diarization when the environment is configured; deterministic mock paths remain available for demos but are not used as performance claims. Alignment assigns transcript material to speaker-time intervals and marks overlap when multiple speaker intervals cover a region. Interruption candidates are generated from temporal relationships and were manually reviewed only at the event level in this project; this yields candidate precision evidence but not recall or F1.

3.5 Conservative RAG-Based Domain-Term Recovery

The RAG module retrieves domain terms from local Markdown and glossary files. Unlike a generic meeting chatbot, it is used only as an auxiliary ASR term-recovery mechanism. The v3 policy is conservative: retrieved candidates can support substitutions for domain terms, but weak evidence is rejected or routed to review. The Earnings-22 experiments evaluate this mechanism on finance-call audio, which supports the term-recovery contribution but not the overlap or diarization contribution.

3.6 Correction Audit and Evidence Provenance

Every correction proposal keeps raw text, proposed text, candidate terms, overlap status, and decision metadata. Controlled overlap-safety and binary safe-to-apply experiments evaluate whether evidence-aware correction decisions reduce unsafe

TalkWeaver: overlap-aware multi-speaker ASR research pipeline

RAG supports domain-term recovery; diarization, overlap, and auditable correction remain the core.

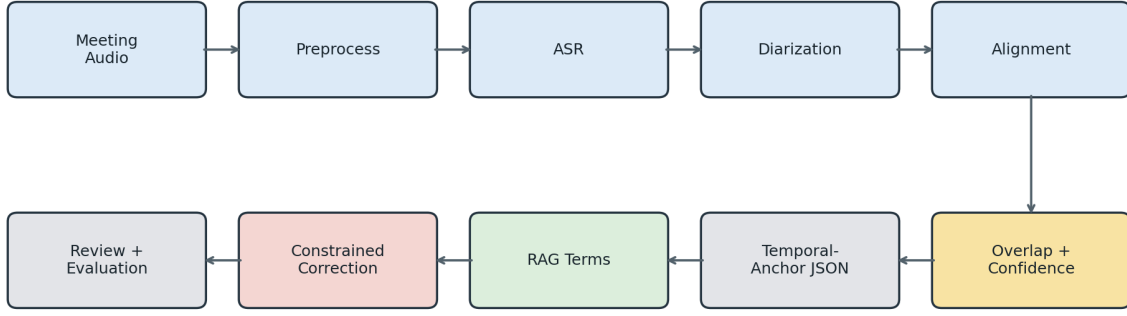


Figure 1: TalkWeaver architecture copied from the repository architecture asset. The paper uses it as a system and evidence-flow diagram, not as a claim that every downstream decision is solved automatically. Exact source paths are listed in the figure/table manifest.

edits in authored stress tests. EvidenceGate adds a learned decision model over audit features, but its leakage audit and independent held-out failure make it a limitation rather than a deployable main result.

4 Experiments

4.1 Evidence Types and Datasets

Table 1 lists the evidence used in this draft. The paper separates real public subsets from controlled text fixtures, proxy mobile measurements, and mock/demo outputs. Mock/demo results are not used for quantitative claims.

4.2 RQ1: How Much Harder Is Meeting Speech?

RQ1 compares read speech and meeting speech under existing ASR and diarization outputs. Table 2 and Figure 2 show that meeting subsets have much higher error rates than the FLEURS read-speech subset. The strongest comparison is Mandarin: FLEURS zh-CN base CER is 0.113, while AISHELL-4 meeting CER is 0.610 in the 12-clip formal subset and 0.537 in the 60-clip subset. The AMI meeting subsets also remain difficult, with base WER 0.398 in the formal subset and 0.349 in the held-out subset; the small model improves AMI held-out WER to 0.299 but is slower.

Diarization results in Table 3 further support the need for evidence-aware maps. AMI diarization has usable DER but only moderate overlap F1. Mandarin AISHELL-4 is substantially harder in this subset, especially for JER and overlap F1.

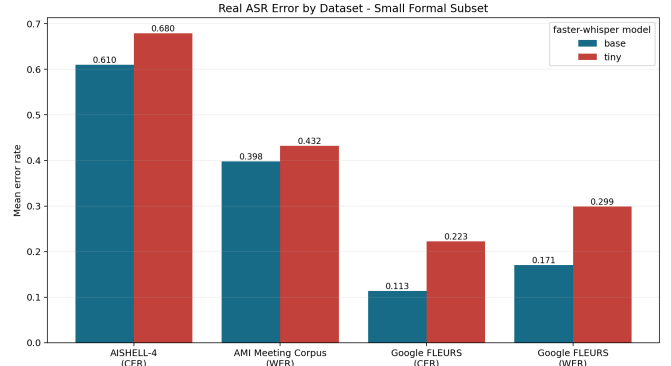


Figure 2: ASR error by dataset, generated from existing public-subset CSVs. The figure supports the meeting-versus-read-speech difficulty claim, not a full benchmark claim.

4.3 RQ2: Can the Pipeline Build Auditable Conversation Maps?

RQ2 evaluates whether TalkWeaver can automatically build speaker-, time-, and overlap-grounded maps from real public audio artifacts. Table 4 summarizes the automatic pyannote workflow on AMI and AISHELL-4. The AMI maps average 4.75 anchors, 4.25 speaker-labeled anchors, 1.79 overlap anchors, and 2.29 needs-review flags per clip. The AISHELL-4 maps average 6.76 anchors, 4.48 speaker-labeled anchors, and 2.66 needs-review flags per scored clip. These are not readability or user-study results; they show that the evidence map is produced end to end on real meeting subsets.

The interruption review is narrower. Ten AMI candidate windows were manually reviewed and all ten were labeled as interruption behavior, giving candidate precision 1.0. Because there is no exhaustive annotated timeline or sampled non-candidate set, recall and F1 are unavailable. Appendix Table 11 records this boundary.

Dataset or artifact	Evidence type	Size	Language	Main use	Repository source
FLEURS formal subset	Real public	30 clips inside 50-clip manifest	en/fr/zh-CN	Read-speech ASR contrast	formal manifest and ASR summary
AMI formal subset	Real public	8 clips	en	Meeting ASR contrast	formal manifest and ASR summary
AMI held-out	Held-out real	24 clips	en	ASR, diarization, automatic map	AMI held-out manifest and summaries
AISHELL-4 60x20	Held-out real	60 clips; 29 scored for diarization/map	zh-CN	Mandarin meeting ASR, diarization, map	AISHELL-4 60x20 manifest and summaries
Earnings-22 v2/v3	Held-out real	12 files v2; 6 files v3	en	Domain-term RAG recovery	Earnings-22 v2/v3 manifests and summaries
Term and overlap fixtures	Controlled	175 term rows; 80 overlap rows	en/fr/zh-CN	Safety stress tests	controlled term and overlap summaries
EvidenceGate held-out	Controlled held-out text	90 authored proposals	multilingual	Leakage and generalization audit	EvidenceGate document and metrics CSV
whisper.cpp Level 1	Local-machine proxy	76 rows	en/fr/zh-CN	Runtime trade-off, not phone deployment	whisper.cpp Level 1 summary

Table 1: Dataset and artifact boundaries. Size and claim level are copied from the final handoff and result files; exact source paths are listed in the claim audit. None of these rows should be described as a full-corpus benchmark.

Dataset	Model	Clips	Metric	Error	Cleaned error	RTF	Source and caveat
FLEURS zh-CN	base	10	CER	0.113	—	0.061	Formal public subset, not full FLEURS
AISHELL-4 formal	base	12	CER	0.610	—	0.050	12-clip Mandarin meeting sanity subset
AISHELL-4 60x20	base	60	CER	0.537	—	0.071	60 clips from 20 recordings, not full AISHELL-4
AISHELL-4 60x20	small	60	CER	0.482	—	0.137	Best CER among tiny/base/small in this subset
AMI formal	base	8	WER	0.398	0.331	0.055	Formal 50-clip manifest row
AMI held-out	base	24	WER	0.349	0.290	0.072	Held-out real subset
AMI held-out	small	24	WER	0.299	0.234	0.176	More accurate but slower than base/tiny

Table 2: ASR evidence for RQ1. Values come from the formal ASR, AMI held-out, AMI held-out small, and AISHELL-4 60x20 summary CSVs listed in the claim audit.

Dataset	Clips	DER	JER	Overlap F1	RTF
AMI held-out	24	0.106	0.307	0.490	0.692
AISHELL-4 60x20	29	0.327	0.713	0.262	0.505

Table 3: Diarization and overlap evidence from the AMI held-out and AISHELL-4 60x20 diarization summary CSVs listed in the claim audit.

4.4 RQ3: Can Conservative RAG Improve Term Recall Without Degrading Quality?

RQ3 uses Earnings-22 because finance calls contain domain terms and are already present as public-audio term-recovery artifacts. Table 5 shows both the positive v3 result and the earlier v2 negative result. On the frozen 6-file v3 subset, base ASR term recall improves from 0.833 to 1.000 with unchanged WER 0.212 for both the evidence-gate and LLM-verifier RAG variants. The term F1 remains 0.833, so the correct claim is term-recall recovery without WER drift, not overall transcript-quality improvement.

The v2 blind subset explains why the method is conservative. On 12 files, RAG improves tiny term F1 from 0.889 to 0.931, but reduces base term F1 from 0.972 to 0.931 and slightly worsens WER. This is a negative result against unrestricted correction and motivates rejecting weak evidence.

Appendix Figure 4 shows the controlled term-rescue chart. It supports mechanism testing on text fixtures, but not public-audio generalization.

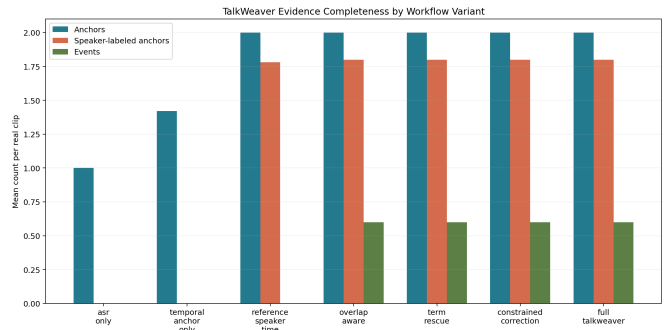


Figure 3: Workflow completeness chart from the repository result charts. It visualizes evidence-map structure and review artifacts; the underlying workflow ablation is reference-assisted in some variants and should not be read as an automatic accuracy result.

4.5 RQ4: What Do Safety and Leakage Analyses Reveal?

RQ4 asks whether automated correction decisions can be trusted. The answer is mixed and is intentionally reported with negative results. Controlled overlap fixtures show that overlap-aware correction can avoid forbidden changes in authored high-overlap cases: overlap-aware rule and LLM variants reach safety pass rate 1.0 on high-overlap groups, while a no-overlap-awareness rule variant has safety pass rate 0.0 and

Dataset	Clips	Pyannote turns	Anchors	Speaker anchors	Overlap anchors	Events	Needs review	Source caveat
AMI held-out	24	4.67	4.75	4.25	1.79	2.04	2.29	Fully automatic map from fixed base ASR and pyannote turns
AISHELL-4 60x20	29	5.66	6.76	4.48	0.38	0.24	2.66	29 clips with usable pyannote turns from the 60-clip subset

Table 4: Automatic conversation-map evidence from the AMI held-out and AISHELL-4 60x20 automatic workflow summary CSVs listed in the claim audit.

Subset	Variant	Model	Rows	WER	Term recall	Term F1	Applied/reviewed	Claim boundary
v3 blind	ASR only	base	6	0.212	0.833	0.833	0/0	Baseline term recall
v3 blind	RAG EvidenceGate v3	base	6	0.212	1.000	0.833	2/1	Conservative term recall gain, WER unchanged
v3 blind	RAG + LLM verifier v3	base	6	0.212	1.000	0.833	2/2	Uses API in 3 rows; same aggregate WER and recall
v3 blind	ASR only	tiny	6	0.252	0.833	0.833	0/0	No v3 improvement for tiny
v2 blind	ASR only	tiny	12	0.222	0.875	0.889	0/0	Weaker model has term room
v2 blind	RAG + LLM verifier v2	tiny	12	0.222	0.958	0.931	2/0	Helps tiny term F1
v2 blind	ASR only	base	12	0.187	0.958	0.972	0/0	Stronger baseline
v2 blind	RAG + LLM verifier v2	base	12	0.187	0.958	0.931	1/4	Negative: lower base term F1

Table 5: Earnings-22 RAG evidence from the v3 blind and final blind v2 summary CSVs listed in the claim audit. RAG is treated as auxiliary domain-term recovery, not general WER improvement.

three forbidden changes in true high-overlap cases. This supports the design principle that overlap should route uncertain edits to audit, but it is not a real-audio correction benchmark.

EvidenceGate is the clearest leakage lesson. An audit-aware grouped test reaches macro F1 1.000, but the repository documents that this uses label-proxy and final-audit fields. A feature audit categorizes 41 fields: 19 allowed pre-decision, 4 risky reference-derived, 14 direct label proxy, and 4 final audit outcome fields. After stricter feature filtering, the evidence-only random forest still reaches grouped-test macro F1 0.976, but on the independent held-out set the best strict model reaches only macro F1 0.325, with needs-review recall 0.033 and unsafe-accept rate 0.167. Table 6 summarizes these results, and Appendix Figures 6–8 provide secondary charts.

4.6 Secondary Track: Multilingual and Local Deployment Trade-offs

The project also includes two required secondary tracks: multilingual evaluation and local deployment trade-off analysis. The multilingual evidence is not a separate claim of cross-lingual robustness; it is used to prevent the method from being described as English-only. The formal ASR summary covers English, French, and Mandarin read speech in FLEURS, plus English AMI and Mandarin AISHELL-4 meeting subsets. In the formal read-speech subset, the base model obtains FLEURS English WER 0.114, French WER 0.227, and Mandarin CER 0.113. In contrast, Mandarin meeting speech remains much harder: AISHELL-4 60x20 base CER is 0.537 and small CER is 0.482. The automatic conversation-map workflow also runs on Mandarin AISHELL-4 scored clips, but this remains a small-subset result rather than a full multilingual benchmark.

The local deployment track is similarly bounded. Appendix Figure 9 and Table 12 report local-machine whisper.cpp Level 1 rows. On AMI, whisper.cpp base has WER 0.351, cleaned

WER 0.280, and RTF 0.056; tiny is faster with RTF 0.031 but less accurate, with WER 0.382 and cleaned WER 0.314. These rows support a speed/accuracy trade-off discussion for local inference, not a phone-device deployment claim. The repository also contains mobile-style proxy artifacts, but the paper treats true mobile deployment as future work.

5 Discussion and Limitations

Meeting difficulty is supported, but not full benchmark performance. The public-data subsets consistently show that meeting speech is harder than read speech. However, the reported results are small subsets: 50 clips in the formal manifest, 24 AMI held-out clips, 60 AISHELL-4 clips, and small Earnings-22 blind subsets. The paper should not claim full-corpus performance or statistical significance.

The conservative workflow is a finding, not a weakness to hide. The safest current real workflow often preserves raw text, adds evidence, and routes risky regions to review rather than changing the transcript. This is consistent with the paper’s framing: evidence-grounded auditing is useful even when correction is not automatically applied. The RAG v3 result is strongest when stated narrowly: base-model term recall improved from 0.833 to 1.000 on the 6-file Earnings-22 v3 subset with WER unchanged at 0.212.

Correction gates are not solved. EvidenceGate demonstrates the risk of leakage. Internal grouped-test results are high, but independent held-out performance is poor, especially for needs-review. Therefore, learned gating should be framed as a negative result and future-work path, not as a deployed component with proven generalization.

Experiment	Evidence type	Key result	Source	Paper-safe interpretation
Overlap safety fixtures	Controlled text	Overlap-aware high-overlap safety pass 1.000; no-overlap rule high-overlap safety pass 0.000 with 3 forbidden changes 41 fields: 19 allowed, 4 risky reference-derived, 14 direct label proxy, 4 final audit outcome	overlap safety summary	Overlap evidence should constrain correction, but this is fixture evidence only
EvidenceGate leakage audit	Controlled audit	Evidence-only random forest macro F1 0.976, false accept 0.000, unsafe accept 0.000	EvidenceGate document	Strong audit-aware scores are leakage-prone
EvidenceGate grouped test	Controlled validation	Best strict macro F1 0.325; needs-review recall 0.033; unsafe accept 0.167	EvidenceGate metrics CSV	Internal grouped validation remains optimistic
EvidenceGate independent held-out	Controlled held-out text	Binary EccoGate macro F1 0.883, unsafe apply 0.053; LLM with evidence macro F1 0.596, unsafe apply 0.324	EvidenceGate metrics CSV	Learned gate is not ready for main-workflow integration
Binary safe-to-apply	Controlled/pilot		binary safety summary	Explicit evidence policy is safer than LLM self-judge in this benchmark, but not a real-audio claim

Table 6: Safety and leakage evidence for RQ4. The table includes controlled and pilot evidence only; it is reported to expose correction risk, not to claim deployed safety.

Interruption and overlap evidence remain limited. Ten reviewed interruption candidates all contained interruption behavior, but this is candidate precision only. There is no exhaustive timeline annotation, no sampled negative set, and no recall/F1. Overlap-aware safety is supported mainly by controlled fixtures; real overlap correction remains open.

Mobile deployment is not complete. The repository includes local-machine and proxy whisper.cpp results, but not a true phone-device deployment. Appendix Table 12 records this boundary.

UI role. The detective-style Streamlit interface is important for inspection and presentation, but it is not the core algorithmic claim. Its role in the paper is to show how evidence maps, audit flags, and retrieved terms can be reviewed.

6 Conclusion

TalkWeaver treats chaotic meeting transcription as an evidence-grounded auditing problem. It builds temporal-anchor conversation maps that preserve ASR text, speaker-time evidence, overlap indicators, retrieved terms, and correction audits. Existing repository evidence supports a focused claim: meeting speech is harder than read speech on the tested public subsets, the automatic map runs on English and Mandarin meeting clips, and conservative RAG can recover domain-term recall on a small blind finance subset without changing WER. The most important limitations are also part of the result: correction gates are leakage-prone, independent held-out generalization is weak, interruption evaluation lacks recall, and true mobile-device deployment remains future work.

A Additional Figures and Tables

This appendix contains secondary charts and tables referenced by the main paper. They are included to preserve traceability while keeping the main paper focused on RQ1–RQ4.

A.1 Controlled Term Recovery

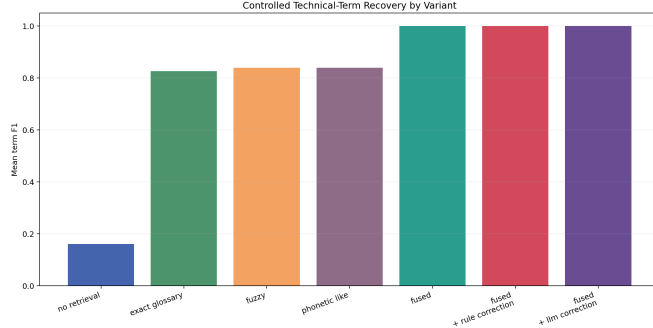


Figure 4: Controlled term-rescue F1 by variant. These are text-fixture results; they support mechanism testing only. Exact source paths are listed in the manifest and claim audit.

Variant	Group	Cases	Term F1	Error after	Review
no retrieval	en easy	7	0.000	0.218	0
fused	en hard	5	1.000	0.351	0
fused+rule	en hard	5	1.000	0.000	0
fused+LLM	en hard	5	1.000	0.223	2
fused+rule	negative control	4	1.000	0.000	4

Table 7: Selected controlled term-rescue rows. The negative-control review count is a safety behavior, not a public-audio performance claim.

A.2 Controlled Overlap Safety

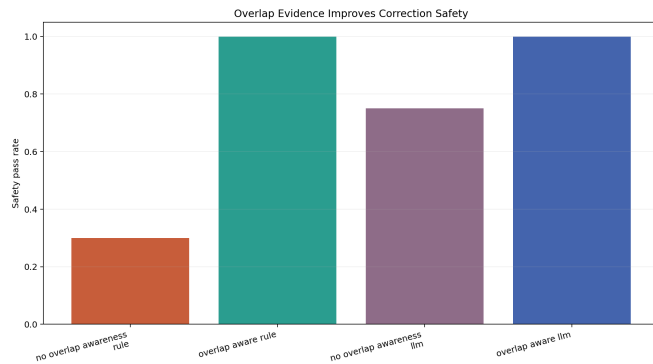


Figure 5: Controlled overlap safety pass rate generated from the controlled overlap summary.

Variant	True overlap	Uncertainty	Cases	Safety pass	Forbidden
overlap-aware rule	true	high	4	1.000	0
overlap-aware LLM	true	high	4	1.000	0
no-overlap rule	true	high	4	0.000	3
no-overlap LLM	true	high	4	0.500	0

Table 8: Selected high-overlap controlled safety rows.

A.3 EvidenceGate Leakage and Held-out Failure

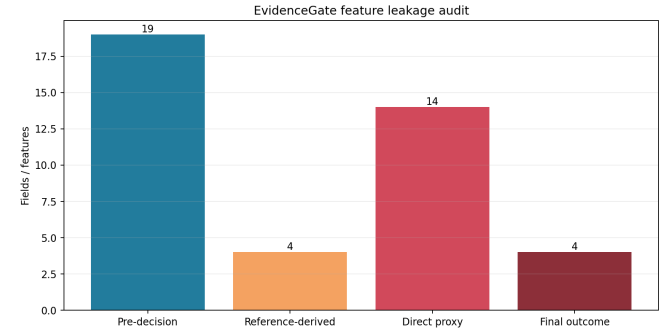


Figure 6: EvidenceGate feature leakage audit. The source document reports 41 audited fields: 19 allowed pre-decision, 4 risky reference-derived, 14 direct label proxy, and 4 final audit outcome fields.

Feature/model	Split	Macro F1	False accept	Unsafe accept	Review recall
audit-aware RF	grouped	1.000	0.000	0.000	1.000
evidence-only RF	grouped	0.976	0.000	0.000	0.962
risk-only RF	grouped	0.924	0.133	0.000	0.769
evidence-only RF	independent	0.325	0.217	0.167	0.033
evidence-only LR	independent	0.316	0.100	0.067	0.000

Table 9: EvidenceGate validation rows from the EvidenceGate metrics CSV and document. The audit-aware grouped result is a policy-distillation sanity check.

A.4 Binary Safe-to-Apply Stress Test

A.5 Interruption Candidate Review

A.6 Runtime and Mobile-Style Evidence

References

- [1] Todo: Verify complete bibliographic entry for aishell-4. Used only as a citation placeholder. Full bibliographic details were not verified from local materials in this draft.
- [2] Todo: Verify complete bibliographic entry for the ami meeting corpus. Used only as a citation placeholder. Full bibliographic details were not verified from local materials in this draft.
- [3] Todo: Verify complete bibliographic entry for the fleurs dataset. Used only as a citation placeholder.

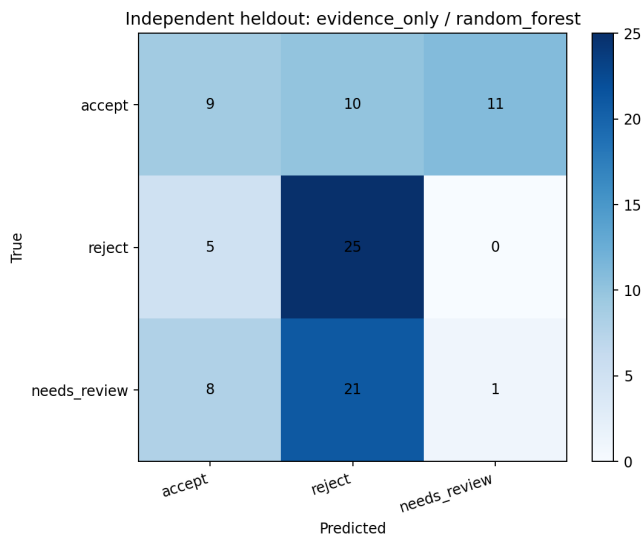


Figure 7: Independent held-out EvidenceGate confusion matrix. The associated metrics show weak generalization, especially for needs-review routing.

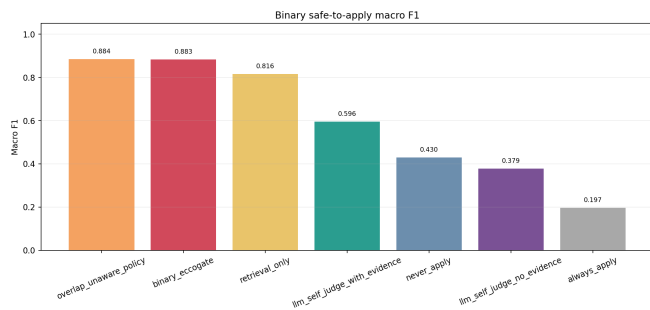


Figure 8: Binary safe-to-apply macro F1. The benchmark is controlled/pilot, not real-audio generalization.

Method	Examples	Macro F1	Unsafe apply	Coverage
always apply	327	0.197	1.000	1.000
retrieval only	327	0.816	0.198	0.385
overlap-unaware policy	327	0.884	0.077	0.272
binary EccoGate	327	0.883	0.053	0.239
LLM judge with evidence	327	0.596	0.324	0.385

Table 10: Selected binary safe-to-apply rows.

Candidates	Reviewed	Interruption	Other labels	Precision	Recall/F1
10	10	10	0	1.000	unavailable

Table 11: Human-reviewed interruption candidate summary. Precision is candidate precision only; recall and F1 are unavailable.

ceedings of *Interspeech*, pages 3754–3758, 2024. Repository source: local paper reading note; arXiv:2401.03506.

Repository evidence mentions Google FLEURS and arXiv:2205.12446, but full author and venue details were not verified from local materials in this draft.

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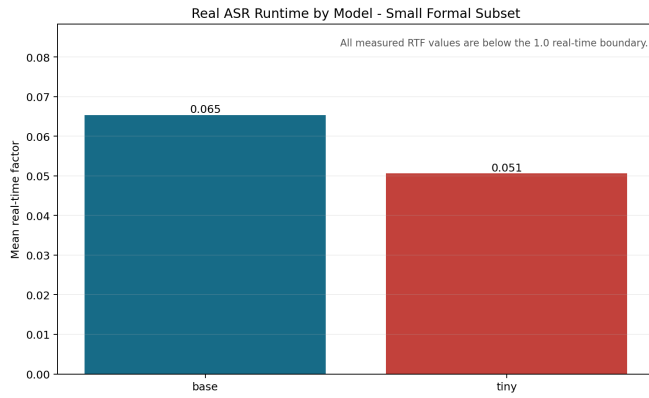


Figure 9: ASR real-time factor chart. This figure supports local runtime trade-off discussion only.

Dataset	Model	Rows	Error	Cleaned error	RTF
AMI	tiny	8	0.382	0.314	0.031
AMI	base	8	0.351	0.280	0.056
FLEURS en	tiny	10	0.186	0.186	0.072
FLEURS en	base	10	0.134	0.134	0.140

Table 12: Selected local-machine whisper.cpp Level 1 rows. These are not phone-device measurements.